

Supplementary Material: Measuring Counseling Interaction Patterns Across Human–Human and Human–LLM Dialogues

Abstract

This document accompanies the main article. It reports the full persona list, complete client- and counselor-side OnCoCo label distributions, hidden-Markov state summaries and state-count sensitivity, transition heatmaps, sentence-embedding projections, per-conversation within-condition variability, the RQ2 multiplicity control, the noise-band placement under each unit of analysis, the classifier-confidence diagnostics, the pre-declared category exclusion sets with their leave-one-out sweep, the unit-of-analysis table, and the prompt-replay experiment of RQ3. Tables and figures are numbered S1, S2, ... and are referenced by those numbers in the main text.

Keywords

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1. Prompts and Case Descriptions

The prompt templates that drove the simulated client are released in the companion repository (directory `prompts/`), one file per persona and template variant, with the conversation history that the platform inserts at generation time replaced by a placeholder. The repository also contains the HH roleplay and H–LLM conversations with personal data masked, the OnCoCo labels aligned to the masked text, the persona definitions, and the segmentation adapter. Prompts of the earliest deployment (Mixtral 8x7B) were not logged by the platform at that time. HH real conversations are not released because they stem from real help-seekers.

The platform assembles each prompt from a persona profile and one of two template families. Plain-text templates open with a role instruction (“# Rollenanweisung” for the first turn, “Du bist ...” for later turns). Chat-template prompts wrap the same persona in the Llama 3 chat format with a structured “## Antwortformat” block. Every logged template carries a brevity instruction at template level (“Maximal 1-2 kurze Sätze”, “1-4 Sätze maximal”, “kurz und präzise”), so the only brevity cue that varies between conversations is the persona profile’s “Sprachliche Merkmale” field (e.g. “Kurze prägnante Rückmeldungen”). Ten of the 26 personas in the analysed H–LLM conversations carry such a cue. Table S18 reports the client-side distance to HH real per prompt configuration, Table S17 the message-length and punctuation statistics behind the verbosity signature, and Table S19 the size-matched subsample check.

2. Persona List

3. Additional Tables and Figures

This appendix reports full label distributions, persistence statistics, within-condition variability, transition heatmaps, and embedding projections.

4. Pre-Declared Category Exclusion Sets

The sensitivity analysis reported in Section 5.5.1 of the main paper removes categories that are arguably fixed by the training setting rather than by counseling behavior. The first two sets below were declared before the analysis was run. S3 is post-hoc and is flagged as such. All three are reproduced here so the choice is auditable.

S1: setting-bound (primary). *Objective of the assignment* (CL-IF-AO-*-Obj-*) and *Extension of the assignment* (CL-IF-AO-*-Ext-*): the assignment is supplied by the persona prompt and the course task, so a simulated client states it because it is in its context window. *Definition of counseling objectives* (CO-IF-AO-*-ICO-*): the counselor mirror of the same construct. *Formalities at the beginning* for both roles (CL-FB-***-*, CO-FA-***-*): the opening formalities

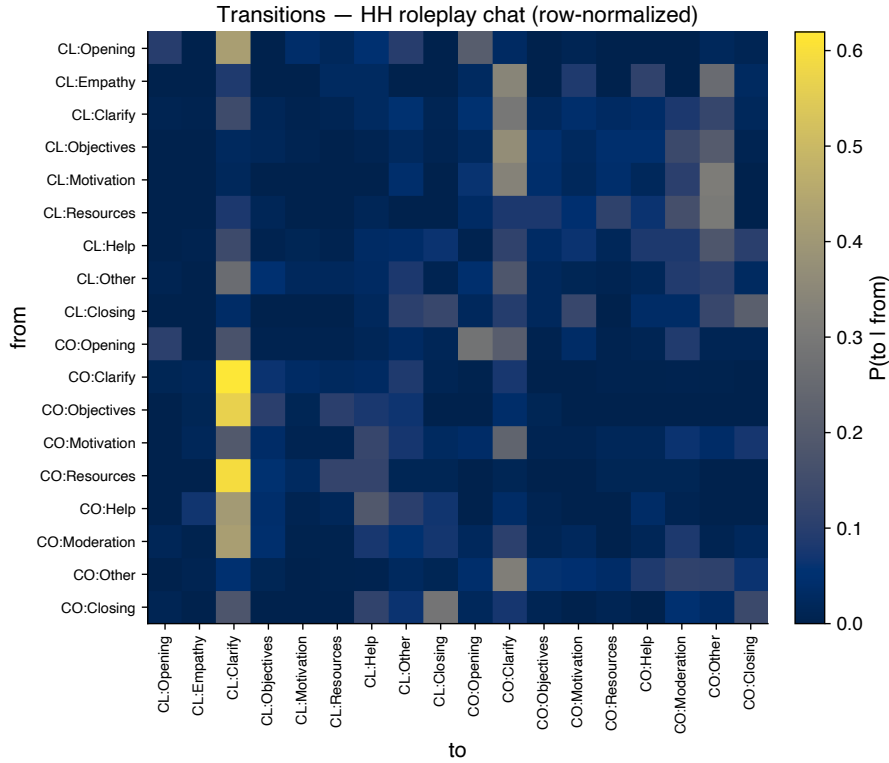


Figure S1. Transition heatmap for HH roleplay (row-normalized).

are platform-dictated. Archived real conversations open with a system join event, and H-LLM sessions open with a scripted first turn. *Inappropriate remark* for both roles (CL-O-***-UCO-*, CO-O-***-UCO-*): excluded not as a setting artifact but as a *classifier* artifact. It carries the lowest median confidence of any label with appreciable support (median 0.48–0.60, against 0.88 or above for every category carrying the behavioral signature). The counselor-side code also absorbs 6.5% of HH real counselor spans against about 1% in the other two conditions. The client-side code is predicted with higher confidence (0.70–0.78) and is excluded only for symmetry with it. This was established in the confidence diagnostics before the present analysis was run.

S2: measurement residual (secondary). *Other statements* for both roles (CL-O-***-O-*, CO-O-***-O-*), the residual category for units the classifier cannot place elsewhere. It is kept as a separate tier because it tests a different threat, namely measurement error rather than a setting confound. Merging the two would let one exclusion set carry two arguments.

S3: closing formalities (tertiary, reported but not pre-declared). *Farewell* for both roles (CO-FC-***-F-*, CL-FC-***-F-*). Session end is partly imposed by the time frame of the exercise, but closing is itself a counseling skill, so the case is weaker and we report rather than assume it.

Deliberately not excluded: Consent (CL-IF-ACP-**-Cons-*). It captures brief acknowledgements and agreement rather than a setting artifact. It is the second-largest client-side category and the largest single contributor to the gap. Removing the largest contributor *because* it is large would be the post-hoc selection the objection warns against. It appears instead as one point among all others in the leave-one-out sweep (Figure S9), where removing it lowers Δ_{CL} by 0.033, from 0.220 to 0.187.

5. Methodological Details Referenced from the Main Text

HMM initialisation. We fit each condition from 20 random initialisations and retain the best-likelihood solution, because the EM likelihood surface for these sequences is strongly multimodal. Across conditions, 20 random initialisations recover 6–11 distinct optima. A single fixed initialisation lands in a markedly inferior one for all three chat conditions (up to $\Delta \log L = 1,468$), and the resulting states then have near-duplicate output distributions. Table S6 reports the sensitivity sweep over $k = 2 \dots 8$, in which the cross-validated optimal state count tracks corpus size rather than any property of the conditions.

Misspecified segmentation. The bottom row of Table S20 is a naive character-class split. It produces 16.2 fragments per message with a mean length of 10 characters, because it splits on a frequent letter rather than only on sentence-final punctuation. The classifier can assign the resulting fragments only to residual categories. Under such a split the between-condition distances

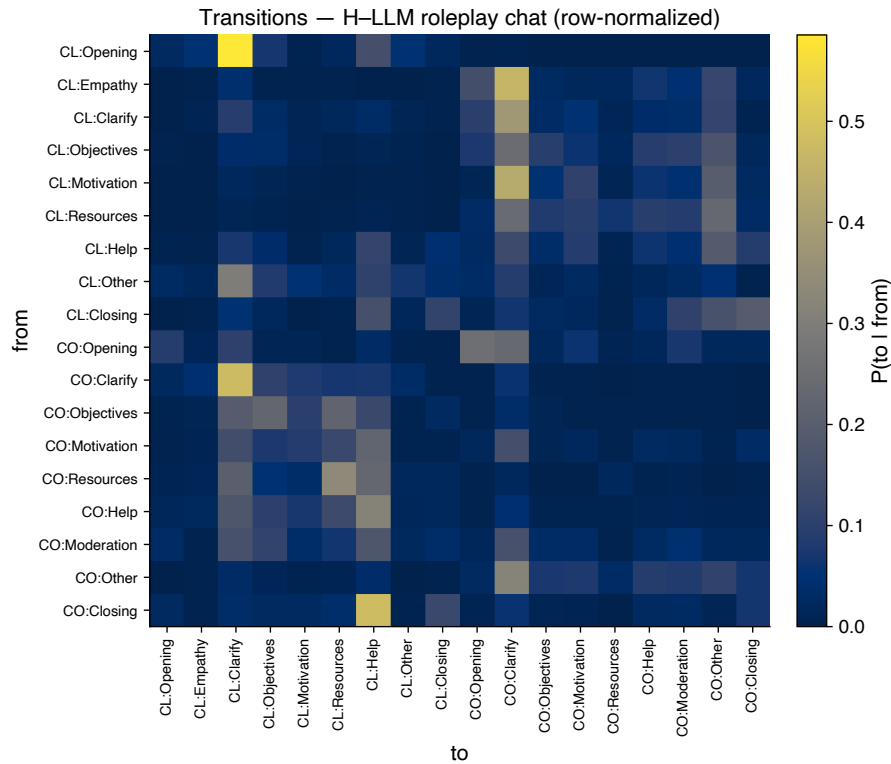


Figure S2. Transition heatmap for H-LLM roleplay (row-normalized).

collapse by a factor of three to four, the client-side gap shrinks to 0.033, a seventh of its value, and the counselor-side gap changes sign. The row is included in the table so that the contrast with a correctly specified sentence split is visible. It is not a segmentation comparison.

Cross-course association (RQ2, secondary). Across all six courses with unambiguous survey linkage, client-side JSD and disagreement with the realism item correlate at Spearman $\rho = .83$ ($p = .042$). The result is fragile. Two of the six courses contribute only 2 and 9 respondents, and dropping them leaves the direction intact but not the significance ($\rho = .80$, $p = .20$ at $k = 4$). A Pearson specification on all six gives $r = .68$ ($p = .13$). We therefore report it as descriptive support for the within-course-line result of Section 5.6 in the main paper rather than as independent evidence.

6. Prompt-Replay Experiment

This section documents the replay experiment of Section 5.7 in the main paper. Its code and prompts are in the companion repository (directories `analysis/prompt-replay/` and `prompts/replay/`). The generated client turns are not released, because the requests carry the unmasked counselor turns of the original conversations.

Design. The platform logged, for every simulated-client turn, the complete prompt it sent to the model, including the persona profile and the conversation history up to that turn. Each turn can therefore be generated again as an independent request whose context is the original history. We regenerate all 1,489 client turns of the 90 GPT-OSS-120B conversations (one turn has no logged prompt and is skipped) with the same model (`openai/gpt-oss-120b`, served with vLLM 0.28 on one A100 80 GB) and the platform’s sampling settings (temperature 1, at most 3,000 tokens, no penalties). The prompt is sent as one user message. Only the instruction block of the prompt varies between arms. The persona block and the history block are byte-identical across arms, which the request builder asserts for every turn. The counselor turns are the original human turns in every arm. This isolates the prompt effect and leaves the counselor’s reaction to a changed client untested.

Arms. Arm O passes the original client text through the current pipeline and is the comparison base. Arm A resends the logged prompt unchanged and tests whether our setup reproduces the deployment. Arm B replaces the instruction block by a generic cue (“write like a real help-seeking person in a counseling chat, realistic, authentic, short”). Arm C replaces it by five rules written against the five signatures of Section 5.2 in the main paper: disclose only what the counselor asked for and at most one new aspect per message, stay with the problem when the counselor proposes something and reject what does not fit, ask the counselor a question in about every third message, do not thank or confirm, and write at most 20 words in chat register without a final period. Arm D keeps C and doses the two rules that overshot: questions become rare (“at most once or twice

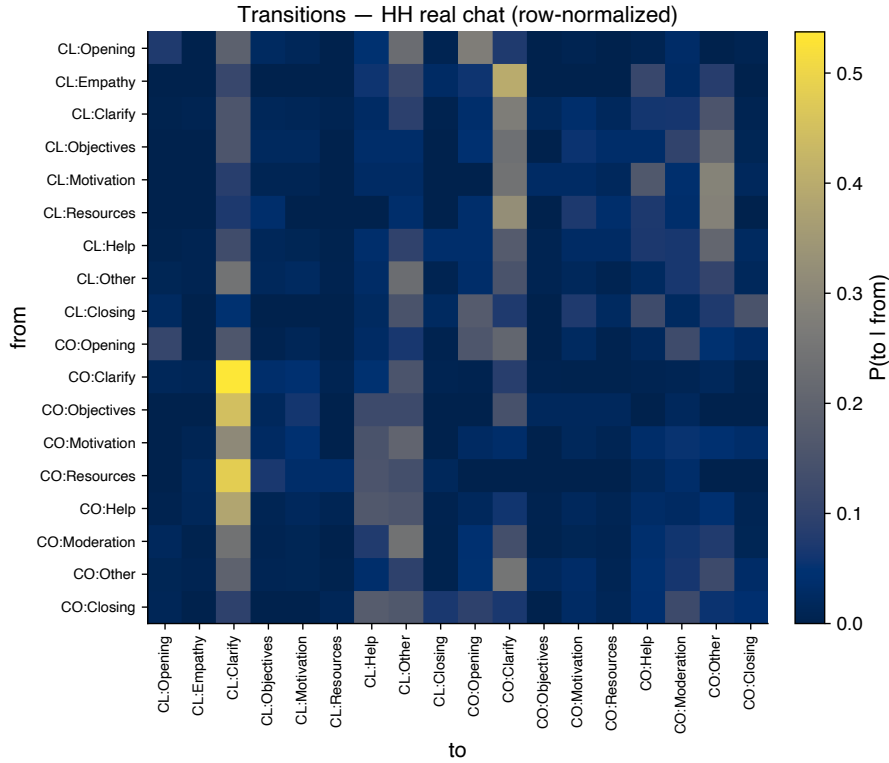


Figure S3. Transition heatmap for HH real (row-normalized).

in the whole conversation”), and short assent is allowed while thanking in every message is not. Arm E keeps D and allows positive feedback about as often as disagreement. Arm F decodes the history before each turn with the HH real phase model of Section 5.3 in the main paper and uses one rule block for the clarification phase (as D, with entry into the help phase left to the counselor) and one for the help phase (feedback on suggestions allowed, no new problem, no planning, closing accepted). Arm G keeps F, removes the form examples from the help-phase block, forbids commitments to implementation (“whether it helps, yes, what you will do with it, no”), and phrases entry into the help phase conditionally instead of “once”.

Measurement. Every arm is converted to the chat format of the corpus and classified with the unchanged pipeline (SaT-6l with the OnCoCo adapter, XLM-RoBERTa-large OnCoCo classifier). The counselor spans are identical across arms, which we verify. The current segmentation reproduces the paper corpus exactly for about 90% of messages, with differences concentrated on emoji and special characters, so arm O and not the paper’s deployment row is the comparison base. The differing messages are split more finely, 1.87 spans per client message in arm O against 1.68 in the deployment, with more one-word spans (14.7% against 12.1%) and more *Other statements* (10.3% against 7.0%). On whole messages, where no segmentation is involved, arm O and the deployment are indistinguishable (JSD 0.003), and on spans arm O lies at 0.058 from the deployment row, inside the deployment’s own split-half band (95th percentile 0.158). Arm A, the logged prompt replayed, lies at 0.083 from the deployment row, also inside that band. All arms pass through the same current segmentation, so differences between arms are not segmentation effects. Table S21 gives every arm on the span and the message unit. Figure S11 gives the decile profiles.

Profiles. Equal distance to HH real does not mean equal behavior. Arms D and G lie 0.233 and 0.213 from HH real but 0.322 and 0.302 from HH roleplay, farther than either lies from real counseling. Arm D matches real clients on the residual *Other statements* (0.232 against 0.241) and on *Consent* (0.182 against 0.171), where role-play clients depart (0.108 and 0.223). It departs where role-play clients do not, with almost no *General positive feedback* (0.004 against 0.047 in HH real and 0.046 in HH roleplay) and few *General requests* (0.007 against 0.027 and 0.040). The dosed arms therefore reach the distance of a human role-play partner by a different route.

Phases. The three-state HMM fitted on HH real (Section 5.3 in the main paper) separates an opening phase (6% of spans), a clarification phase (80%), and a help-and-closing phase (15%). Every condition is decoded with this one reference model, because states of separately fitted models are not comparable: the third state of the H-LLM fit is a solution-planning mode (Help 0.27, Resources 0.10, Objectives 0.11) that occupies 39% of its spans, while the third state of HH real is a help-and-closing phase. Under the reference model, real clients express problems, questions, and disagreement in the clarification phase and positive feedback almost only in the help phase (0.241 of their spans there against 0.026 in clarification). Real clients initiate

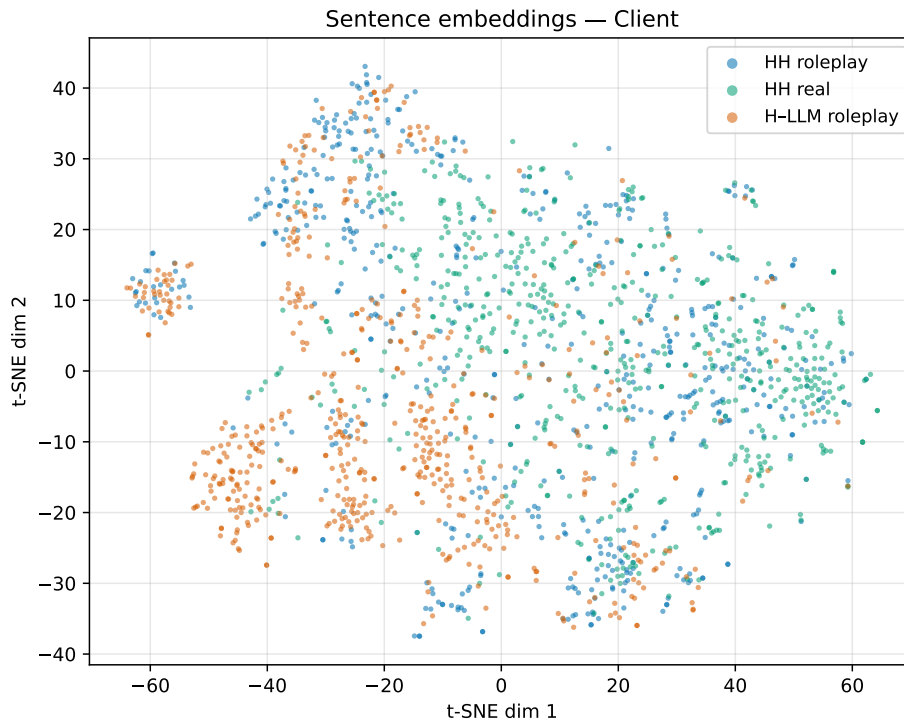


Figure S4. t-SNE projection of sentence embeddings by condition (Client).

44% of the moves from clarification into help. The turn-level test decodes the original history before each client turn and again with the client’s spans appended. A move from clarification to help counts as client-led entry. Table S22 reports the rates.

Example echo. Arm F copied the form examples of its help-phase block. 105 of its 243 help-phase turns read “ja das hilft danke”, so part of its positive feedback was copied rather than generated. Arm G removed the examples, and its 243 help-phase turns contain 227 distinct messages. The main text reports G, not F.

7. Human Validation on the Study Corpora

This section reports the expert validation of the OnCoCo classifier referenced in Sections 4.2, 5.5.2, and 7 of the main paper.

Setup. Both human corpora carry expert annotation under the OnCoCo scheme, produced by two different annotation teams: 4,354 units in 64 HH roleplay conversations and 3,591 chat units in 54 HH real conversations, the latter exported from MAXQDA and reconstructed from the source documents paragraph by paragraph. The experts segment by function and the pipeline by sentence, and the two cuts rarely coincide (boundary F1 0.13 on HH roleplay). We therefore re-classified the experts’ own units with the pipeline’s role prefix, so that the comparison holds the segmentation fixed and only the labels differ. No simulated-client conversation has expert annotation.

Agreement. Table S23 gives unit-level agreement per corpus and role. Both corpora lie far below the in-domain figures of the OnCoCo test set (accuracy 0.79, model-to-human $\kappa=0.88$) and below the human intercoder agreement of the scheme ($\kappa=0.84$). HH real is the harder corpus throughout, with $\kappa=0.34$ against 0.45, and on its client side the classifier does not beat the majority baseline, because *Problem statement* makes up 39% of the units there and is recalled at 0.22. At the level of the scheme’s pillars both corpora reach 0.81 to 0.82, so the broad field of action is recognized and the fine category is not. The classifier is too generous rather than blind: on the counselor side it uses 50 categories where the annotators use 33 to 35, and every category the annotators never use is by construction an error. The single largest confusion is *Problem statement* against *Problem definition*, 197 units in both directions in HH real and 90 in HH roleplay, with *Own emotional expression* as the third member of the same family, and merging only these three labels raises the client-side κ from 0.48 to 0.55 on HH roleplay and from 0.33 to 0.45 on HH real. A tolerant reading that merges neighbouring categories (backchannel against residual, consent against positive feedback, reflection types, question types, and the problem family) raises κ to 0.62 on HH roleplay and 0.54 on HH real. That is the most generous figure obtainable, not the strictest, and every analysis that reports single categories is bound by the strict one.

Signature categories. Table S24 gives precision, recall, and F1 for the categories that carry the signatures of Section 5.2 in the main paper, next to the shares the experts and the classifier assign. *Consent*, *General request*, and *Rejection* are the

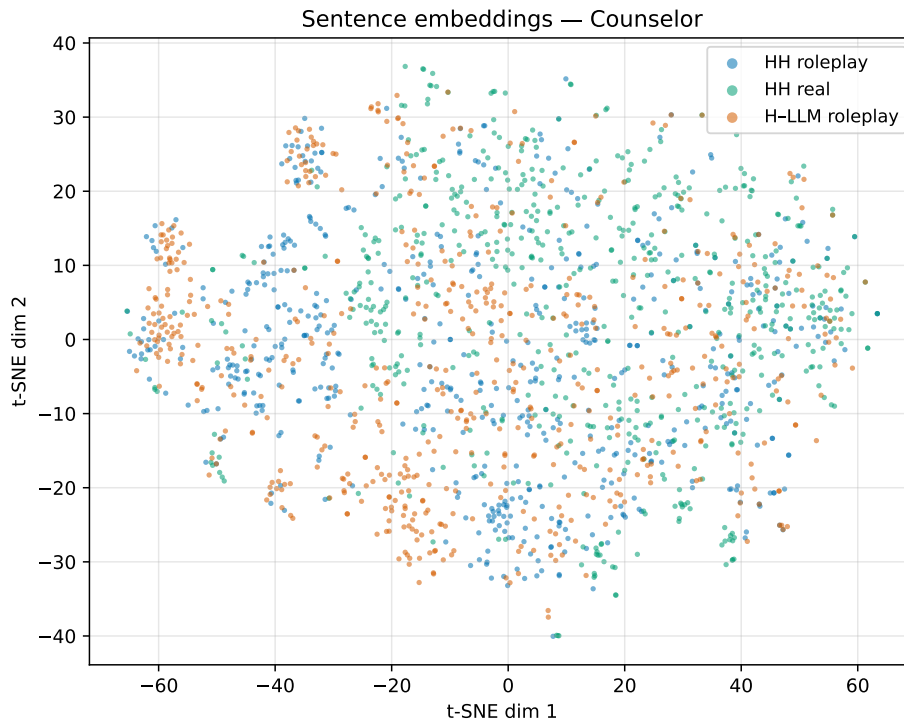


Figure S5. t-SNE projection of sentence embeddings by condition (Counselor).

best-identified categories in both corpora. The largest error is the under-detection of *Problem statement* in HH real (recall 0.22, expert share 0.202 against 0.068 for the classifier). Table S25 pools the categories into the three client-side buckets of Section 5.2. Reciprocal requests are measured almost without loss (0.94 of the expert share in both corpora), solution-space engagement in the same direction with a rough match of the shares, and problem disclosure is over-reported by 15% in HH roleplay and under-reported by 34% in HH real. The units the classifier loses from the problem bucket do not move within the bucket but leave it, mostly to *Consent*, the residual categories, and *Own emotional expression*. Short units carry most of this loss (recall 0.09 at one to three words against 0.69 at 26 words and more), but length explains only a sixth of the gap between the corpora. The classifier is worse on real counseling in every length class.

Distance compression. With expert labels on both human corpora, the distance between them can be computed twice on identical units. Under expert labels the client-side JSD between HH roleplay and HH real is 0.359 [0.333, 0.402], under classifier labels 0.196 [0.182, 0.244], and the pipeline’s own SaT spans give 0.233, so the unit is not the cause. A paired conversation-level bootstrap puts the compression at 0.163 [0.113, 0.197] on the client side and 0.140 [0.107, 0.160] over both roles. Within HH real, where one team and one codebook apply, the same comparison between the three regional services shows no compression (0.012 to 0.056, all intervals including zero). Two readings remain open: the classifier compresses exactly this contrast, or part of the expert distance is a codebook and team difference between the two annotation groups. We therefore read the result as a direction and not as a clean attenuation estimate. In either reading the classifier measures the distance between the two human conditions smaller than the experts do. Whether the same holds for the distance to simulated clients is the open question below.

What the validation cannot settle. The H-LLM term of Δ has no expert labels. If the classifier compresses the distance from real counseling to simulated clients by the same amount as the distance to human roleplay, Δ_{CL} is unchanged. If it compresses it less, because simulated text is closer to the register of the training material, Δ_{CL} shrinks. A stratified sample of about 400 expert-coded simulated-client units would close this gap.

Repair attempts. Three changes to how the classifier is fed do not repair the loss on HH real. Prefixing the preceding counselor turn collapses accuracy to 0.08, because 66% of the predictions then come from the other role’s inventory. Prefixing the client’s own preceding unit raises problem recall from 0.47 to 0.53 but lowers accuracy from 0.36 to 0.26, because the model copies the previous label. Merging units of at most three words into the following unit moves the recall from 0.47 to 0.48 and the between-corpus contrast by 0.001. The bias is a property of the model, and a fix would need retraining with a context field.

Inappropriate remark. The category is a lexical artifact. Apology formulas (“tut mir leid”), prescriptive phrasing (“du musst”), absolute words (“niemand”), and generalizations trigger it at 6 to 10 times the base rate, and the four strongest triggers

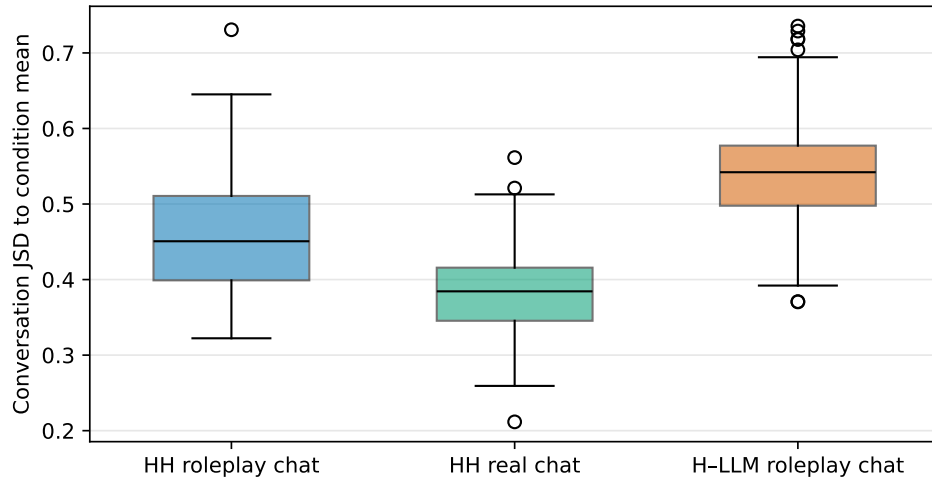


Figure S6. Within-condition variability (Client): distribution of conversation-level JSD to the condition mean.

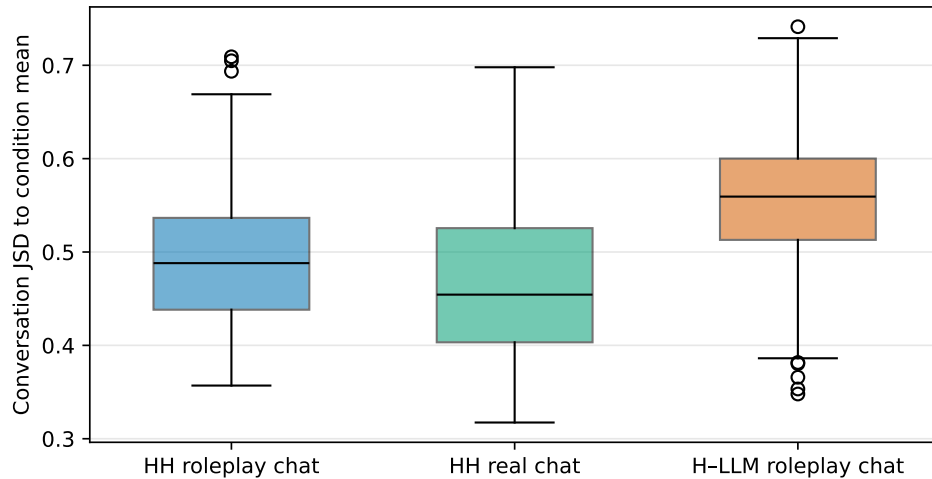


Figure S7. Within-condition variability (Counselor): distribution of conversation-level JSD to the condition mean.

occur in none of the 49 units the experts themselves coded as inappropriate. The classifier marks assertive counselor speech, that is assessments, advice, and reassurance delivered in an informal register, as inappropriate, which is why the primary exclusion set of Section S4 removes the category.

8. Weighting and Persona-Matched Checks

Two further checks on the client-side proximity gap of Section 5.1 in the main paper (Table S26). Pooled label proportions weight conversations by their length. Weighting every conversation equally instead, by averaging the per-conversation label distributions with explicit zeros for unused labels, moves Δ_{CL} from 0.220 to 0.217. Restricting both roleplay conditions to the six personas that occur in HH roleplay and in H-LLM (40 and 220 conversations, persona identity as in Section S2) moves it to 0.208, and equal weighting on the matched set gives 0.209. All intervals are conversation-level bootstraps with 5,000 resamples. Script: `analysis/semantic/oncoco/oncoco_equal_weight_persona_delta.py`.

9. Repeated Participation and Absolute Turn Positions

This section checks two assumptions behind the intervals of Section 3.1 and the temporal signatures of Section 5.2 in the main paper.

Repeated participation. The 414 H-LLM conversations come from 281 students, identified by the platform's course-member account within each export. 202 students held one conversation, 39 two, 28 three, 10 four, and 2 five, and no student appears in more than one course or model. The main paper resamples conversations. Table S27 resamples students instead, so that

Behavioral signatures across model generations (Client)

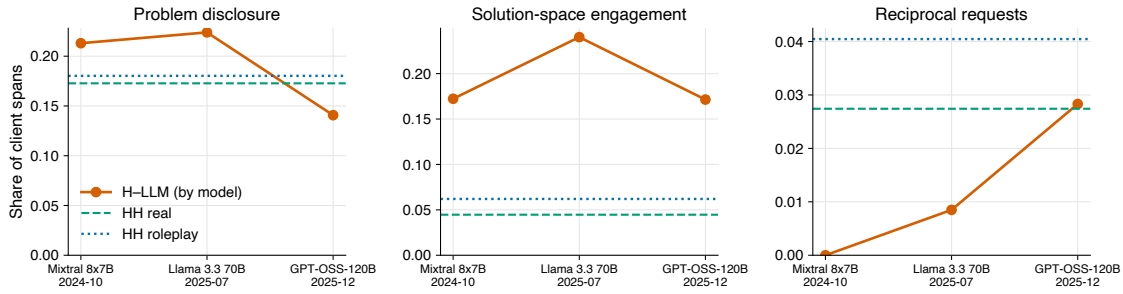


Figure S8. Behavioral signatures across model generations (share of Client spans), with human baselines (dashed for HH real, dotted for HH roleplay). Reciprocal questioning and restraint in problem disclosure improve for the newest model, but solution-space drift persists across generations.

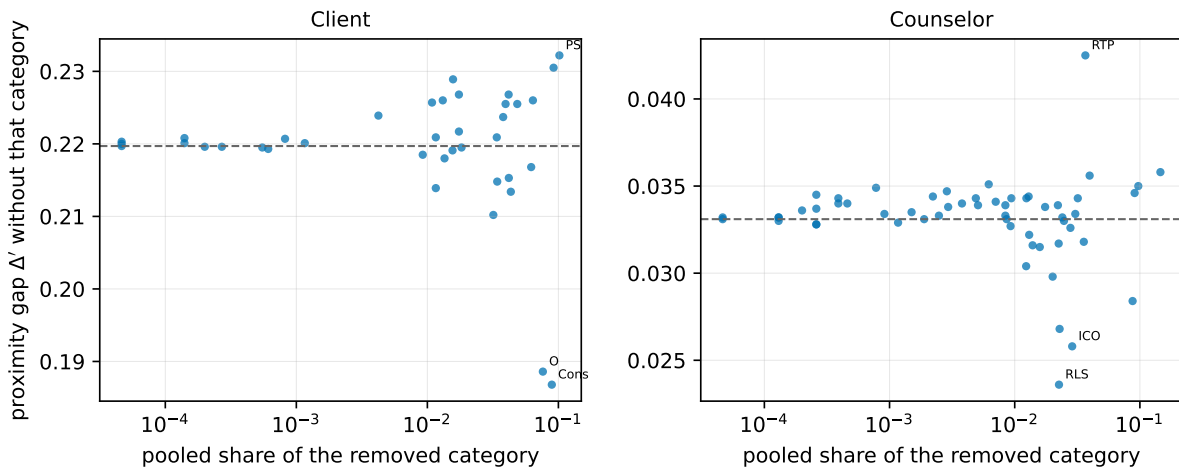


Figure S9. Leave-one-out sweep: the proximity gap Δ recomputed with each category removed in turn, plotted against that category's pooled share. The dashed line is Δ on the full support. No single removal brings the client-side Δ below 0.187. The two categories whose removal lowers it most (*Consent* and *Other statements*) each take off about 0.03.

all conversations of a student enter or leave a bootstrap sample together. HH real and HH roleplay are resampled by conversation in both columns, because the HH roleplays ran under a shared course account and repeated participation cannot be reconstructed there. Resampling students moves no interval bound by more than 0.004, and every per-model comparison of Figure 2 in the main paper keeps its reading. Script: `analysis/semantic/uncoco/uncoco_student_clustered_bootstrap.py`.

Absolute turn positions. The deciles of Section 5.2 in the main paper place each client span relative to the client's own spans in that conversation. Real conversations are longer, with a median of 28 client turns against 18 in HH roleplay and 16 in H-LLM, so a decile covers more turns in HH real. Table S28 therefore counts client turns in absolute terms, from the start and from the end of the conversation. A client turn is a maximal run of consecutive client messages, because the HH real export merges consecutive lines of one speaker while the other conditions keep them apart. Every signature holds. In the first three client turns, problem labels take 0.319 [0.302, 0.336] of H-LLM client spans against 0.122 [0.088, 0.158] in HH real and 0.173 [0.132, 0.216] in HH roleplay. In the last three client turns, solution-space labels take 0.368 [0.339, 0.396] against 0.029 [0.013, 0.046] and 0.094 [0.061, 0.132]. From the fourth client turn on, the H-LLM solution share is three to nine times the HH real share at the same turn, and it keeps rising in the 121 simulated conversations that run beyond 20 client turns. The H-LLM *Rejection* share stays at or below 0.012 in every bin. Intervals are conversation-level bootstraps. Script: `analysis/semantic/uncoco/uncoco_absolute_turn_profile.py`.

10. Held-out Replay

This section tests whether the dosed prompts of Section 5.7 in the main paper transfer to conversations they were not written for. Arms D and G were written and dosed on the 90 GPT-OSS-120B conversations. We applied both unchanged to the

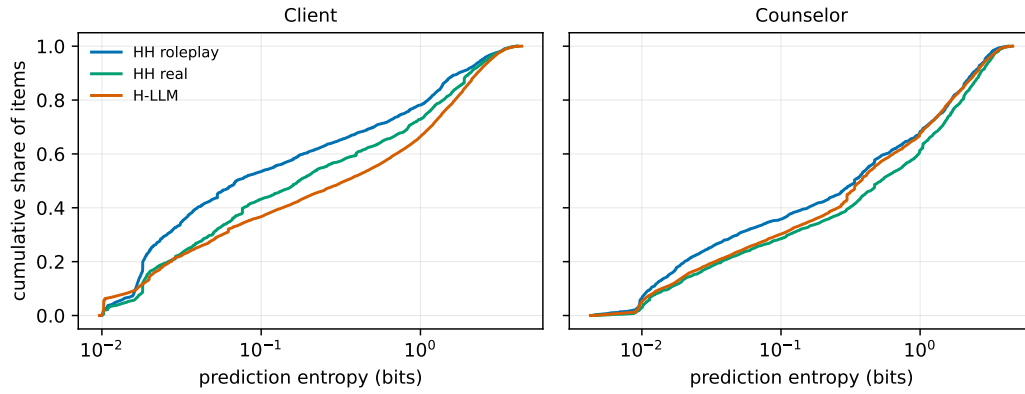


Figure S10. Cumulative distribution of prediction entropy by condition and role (symlog x -axis). HH real carries the heaviest uncertainty tail on both roles. The pooled client-side confidence ranking in the main text is driven by span length, which the length-matched comparison controls.

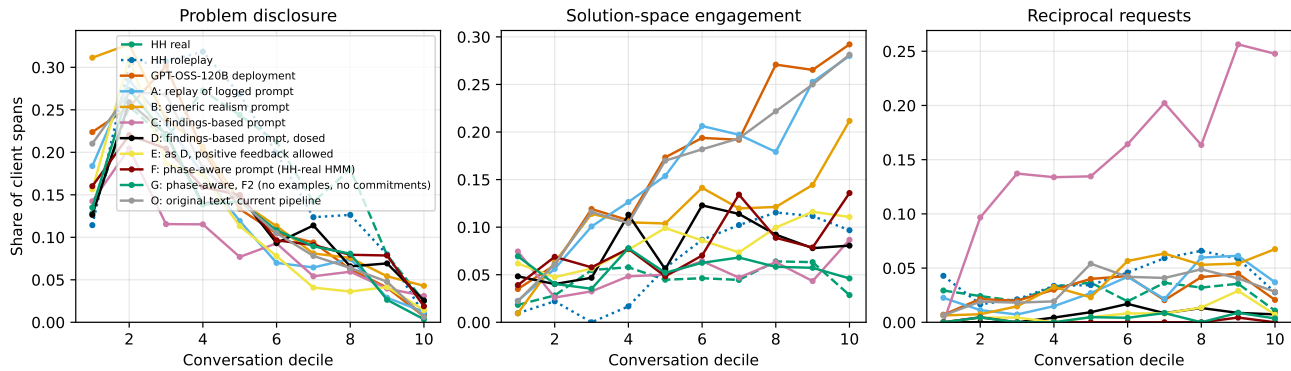


Figure S11. Client-side decile profiles of the replay arms next to the human conditions: share of client spans on problem disclosure, solution-space engagement, and general requests per conversation decile.

117 Llama 3.3 70B conversations that ran under the same plain-text template family and passed the quality filter. 114 of them use personas that never occur in the GPT-OSS deployment. GPT-OSS-120B generates every arm with the settings of Section S6, so arm A, the logged Llama prompt replayed by GPT-OSS-120B, is the baseline that isolates the prompt. Arm O passes the original Llama client text through the current pipeline. The arms were classified on a GPU node with the package versions of Table S30, and that setup reproduces the local classification of arm O exactly (3,561 of 3,561 messages with identical spans and 5,417 of 5,417 identical span labels). Scripts: `analysis/prompt_replay/` and `analysis/semantic/ncoco/ncoco_prompt_replay_heldout.py`.

Both prompts transfer (Table S29). The dosed rules lower the distance to HH real from 0.421 under the logged prompt to 0.287, and the phase-dependent rules to 0.253. Their proximity gaps are 0.054 [0.010, 0.089] and 0.020 [−0.024, 0.056], so the phase-dependent arm reaches the role-play level and the dosed arm stays slightly above it. The reduction against the logged prompt is larger than on the development conversations (0.134 and 0.169 against 0.093 and 0.112), where the logged prompt started closer to real counseling (0.326 against 0.421). The residual departures of the development set return. Positive feedback stays too rare (negative-to-positive ratios 7.32 and 2.15 against 0.78), early disclosure falls below the real level (0.074 and 0.095 against 0.128), and client-led entry into the help phase stays below the real-client rate (1.6% and 0.7% against 2.0%). Both arms again lie farther from HH roleplay (0.327 and 0.309) than from HH real, so they reach the distance of a role-play partner by a different route, as on the development conversations.

11. Pipeline and Generation Parameters

Table S30 lists the settings behind every number in the main paper. Code, prompts, and the adapter are in the companion repository.

12. Client Sentiment Against Both Human Baselines

This section reports the client sentiment behind the recommendation in Section 6 of the main paper. We scored every client message with the German sentiment model `oliverguhr/german-sentiment-bert`, the classifier that Träg et al. (2026) used for AI-guided debriefings. Table S31 gives the shares per condition. Roleplay clients are about as positive as simulated clients. Both are more positive than real clients. We compared the positive share per conversation with two-sided Mann–Whitney tests. HH roleplay differs from HH real (Cliff’s $\delta=0.35$, $p=.001$). H–LLM differs from HH real as well ($\delta=0.29$, $p < .001$). H–LLM and HH roleplay do not differ ($\delta=0.05$, $p=.50$).

Table S1. Personas used in chat roleplays (short descriptions truncated to the first sentence).

#	Persona name(s)	Conv.	Case brief (first sentence)
1	Luisa	89	Luisa ist genervt von den Streitereien ihrer Eltern.
2	Michael Schlosser	70	Herr Schlosser möchte gerne wieder eine Arbeit finden.
3	Elke	61	Elke macht sich Sorgen um ihren Sohn.
4	Jessica Bergmann	53	Jessica ist sich unsicher.
5	Marlene Grafe	43	Sie sucht eine besser bezahlte Stelle im Hauswirtschaftsbereich oder in der Gastronomie in Teilzeit (30 Stunden).
6	Marco	29	Marco hat Schwierigkeiten, sich selbst zu organisieren.
7	Eddie, Jakob, Lina, Susi, Tanja Müller	28	Lina befindet sich derzeit im Entzug.
8	Frau Wolke	25	Frau Wolkes Sohn Jan ist 14 Jahre alt und wohnt zuhause bei ihr.
9	Eddie	19	Eddie macht sich Sorgen um seinen Sohn.
10	Frau Schuster	13	Frau Schusters Sohn Max ist 8 Jahre alt geht zur Schule und ist oft anschließend im Hort.
11	Michaela	9	Verbesserung der Kommunikation innerhalb der Familie
12	Regina	6	Persona arbeitet 20 Jahre im gleichen Job und kann nicht mehr die Tätigkeit ausführen, hat körperliche Beschwerden, wenn sie nur an die Arbeit denkt; möchte\ldots
13	Lars	6	Lars befindet sich derzeit im Entzug.
14	Boris	5	Familienberatung suchen, um eine neue Perspektive auf familiäre Konflikte zu gewinnen.
15	Lara	4	Lara "funktioniert zwar nach außen hin gut", fühlt sich aber innerlich unter Druck und überfordert.
16	Mira	4	Mira fühlt sich zunehmend emotional erschöpft und überfordert im Studium.
17	Leyla	3	Leyla steht kurz nach dem Abitur und ist unsicher, welchen beruflichen Weg sie einschlagen möchte.
18	Annika	3	Annika kommt oft erst spät abends nach Hause in ihre 2-Zimmer-Wohnung in Berlin.
19	Michael Müller	2	Bewältigung der emotionalen Auswirkungen der Scheidung
20	Ute Meier	2	Unterstützung bei der Organisation eines stabilen Alltags nach der Scheidung.
21	Robert 12	2	Probleme des Soiihnen
22	Charlie	1	"Ich weiß nicht mehr weiter ... es macht einfach keinen Sinn mehr." Charlie hat ihr Mediendesign-Studium vor wenigen Monaten abgebrochen.
23	Elias	1	Elias hat das Gefühl, dass sein aktuelles Studium keine sichere Perspektive ist – er fühlt sich unter Druck sich gegen die Konkurrenz durchzusetzen und hat\ldots
24	Jonas	1	Jonas steht kurz vor dem Studienabschluss – er muss noch ein Seminar absolvieren und seine Bachelorarbeit schreiben.
25	Marie	1	Trauer
26	Lars	1	Unterstützung bei der Überwindung der Glücksspielsucht
27	Tobias Meier	1	Tobias sucht Unterstützung bei seiner Alkoholsucht.

Table S2. Persistence of top label shifts across conversations. Med. diff is the difference in *per-conversation median* label proportion, unweighted across conversations. It can diverge from the pooled shifts reported for RQ1 in the main paper (Section 5.1) when conversations contribute few spans. Conversations that never use a label contribute zero rather than being dropped. AUC is the more reliable, rank-based summary. It gives the probability that a random conversation from condition A exceeds one from condition B. $P(a>med)$ is the fraction of condition-A conversations above the condition-B median.

Comparison	Speaker	Label	Med. diff	AUC	$P(a>med)$
HH roleplay vs H-LLM roleplay	Client	Client: Consent	+0.186	0.925	0.971
HH roleplay vs H-LLM roleplay	Client	Client: Other statements	+0.103	0.837	0.882
HH roleplay vs H-LLM roleplay	Client	Client: Report on the implementation of recommendations for action	-0.045	0.249	0.059
HH roleplay vs H-LLM roleplay	Client	Client: Articulation of reasons for a change in the client	-0.037	0.298	0.103
HH roleplay vs H-LLM roleplay	Client	Client: Objective of the assignment	-0.026	0.346	0.294
HH roleplay vs H-LLM roleplay	Counselor	Counselor: Moderation	+0.067	0.714	0.824
HH roleplay vs H-LLM roleplay	Counselor	Counselor: Professional situation	+0.000	0.352	0.059
HH roleplay vs H-LLM roleplay	Counselor	Counselor: Request for concerns	+0.014	0.632	0.647
HH roleplay vs H-LLM roleplay	Counselor	Counselor: Complex Reflection	-0.024	0.419	0.368
HH roleplay vs H-LLM roleplay	Counselor	Counselor: Definition of counseling objectives	+0.000	0.435	0.485
HH real vs HH roleplay	Client	Client: Other statements	+0.112	0.880	0.981
HH real vs HH roleplay	Client	Client: Consent	-0.062	0.377	0.245
HH real vs HH roleplay	Client	Client: Objective of the assignment	-0.033	0.347	0.226
HH real vs HH roleplay	Client	Client: Formalities for conclusion	-0.034	0.249	0.094
HH real vs HH roleplay	Client	Client: Feedback on previous attempts at solutions	-0.018	0.401	0.264
HH real vs HH roleplay	Counselor	Counselor: Inappropriate remark	+0.039	0.757	0.717
HH real vs HH roleplay	Counselor	Counselor: Targeted, precise request	+0.054	0.755	0.774
HH real vs HH roleplay	Counselor	Counselor: Request for concerns	-0.009	0.419	0.377
HH real vs HH roleplay	Counselor	Counselor: Simple reflection	-0.016	0.443	0.415
HH real vs HH roleplay	Counselor	Counselor: Offer to use the counselor's professional resources	-0.031	0.295	0.075

Table S3. Full client-side OnCoCo label distribution (proportions) for the three chat conditions. The table lists labels reaching ≥ 0.01 in any condition.

Label	HH rp	HH real	H-LLM
Problem statement	0.083	0.087	0.109
Problem definition	0.097	0.085	0.093
Objective of the assignment	0.039	0.016	0.078
Formalities at the beginning and greeting	0.054	0.047	0.070
Report on the implementation of recommendations for action	0.010	0.012	0.057
Considering resource activation at a professional level	0.023	0.007	0.054
Articulation of reasons for a change in the client	0.012	0.025	0.052
General positive feedback	0.046	0.047	0.049
Positive feedback on specific recommendations for action	0.015	0.008	0.044
Consent	0.223	0.171	0.044
Feedback on previous attempts at solutions	0.042	0.023	0.043
Considering resource activation at the level of friends and family	0.009	0.004	0.043
Extension of the assignment	0.019	0.018	0.041
Disclosure of personal data	0.035	0.029	0.041
Other statements	0.108	0.241	0.029
Empathy for third parties or related to the present situation (PT)	0.006	0.006	0.020
Eliciting/Evoking "Change-Talk" (MI)	0.014	0.014	0.019
Final success	0.006	0.004	0.017
Own emotional expression	0.012	0.030	0.015
Further use of professional resources of counselor	0.018	0.004	0.015
Concern for another person (EC)	0.010	0.003	0.014
Formalities for conclusion	0.033	0.011	0.014
General request	0.040	0.027	0.012
Negative feedback on specific action recommendation	0.004	0.014	0.011
Inappropriate remark	0.013	0.021	0.006
Rejection	0.022	0.029	0.005

Table S4. Full counselor-side OnCoCo label distribution (proportions) for the three chat conditions. The table lists labels reaching ≥ 0.01 in any condition.

Label	HH rp	HH real	H-LLM
Other statements	0.146	0.145	0.146
Simple reflection	0.101	0.082	0.100
Formalities at the beginning/Opening of the conversation	0.102	0.088	0.089
Moderation	0.122	0.107	0.077
Complex Reflection	0.025	0.024	0.040
Request for concerns	0.060	0.039	0.036
Definition of counseling objectives	0.021	0.007	0.035
Thanks and appreciation to the client	0.036	0.021	0.034
Request for previous attempts at solutions	0.032	0.018	0.033
Targeted, precise request	0.025	0.067	0.032
Professional situation	0.002	0.005	0.030
Offer to use the counselor's professional resources	0.026	0.009	0.028
Professional, formal, institutional recommendation	0.016	0.005	0.028
Request for emotional state	0.030	0.036	0.026
Social relationships	0.021	0.019	0.026
Request for objectives/wishes	0.017	0.013	0.025
Farewell	0.031	0.022	0.021
Articulation of the perceived ability to change in the client (MI)	0.011	0.008	0.018
Request for personal data	0.023	0.037	0.016
Eliciting/Evoking "Change-Talk" (MI)	0.006	0.010	0.014
Advice	0.015	0.019	0.011
Health (mental and physical)	0.022	0.019	0.011
Evaluation, Interpretation	0.010	0.026	0.011
Leisure	0.005	0.009	0.010
Suggestion for resource activation at professional level	0.019	0.020	0.010
Inappropriate remark	0.011	0.065	0.008
Wish	0.010	0.014	0.008

Table S5. HMM state summaries (three states per condition, best of 20 random initialisations). States are ordered by occupancy (Occ.), the share of spans assigned to a state, so state 0 is the most frequent.

Condition	State	Occ.	Top labels (prob.)
HH roleplay	0	0.73	Clarify (0.57), Other (0.14), Moderation (0.07)
	1	0.19	Closing (0.25), Help (0.21), Other (0.17)
	2	0.08	Opening (0.69), Clarify (0.23), Other (0.04)
HH real	0	0.80	Clarify (0.53), Other (0.22), Help (0.07)
	1	0.15	Other (0.32), Help (0.20), Clarify (0.11)
	2	0.06	Opening (0.65), Other (0.14), Clarify (0.14)
H-LLM roleplay	0	0.50	Clarify (0.58), Other (0.09), Objectives (0.09)
	1	0.39	Help (0.27), Clarify (0.13), Other (0.12)
	2	0.10	Opening (0.54), Clarify (0.30), Objectives (0.04)

Table S6. HMM state-count sensitivity. Δ BIC is relative to each condition's best value (bold, $\Delta = 0$). "Optimal k " gives the state count minimising BIC and the one maximising 5-fold cross-validated held-out log-likelihood (conversation-level folds). Each cell is the best of 20 random initialisations. The cross-validated optimum grows with corpus size (4, 6 and 8 states). We therefore hold $k = 3$ fixed across conditions. The BIC optimum is not monotone in corpus size.

Condition	Spans	Δ BIC by state count k							Optimal k	
		2	3	4	5	6	7	8	BIC	CV
HH roleplay	3,927	469	70	0	53	187	262	438	4	4
HH real	4,927	302	0	12	112	260	403	502	3	6
H-LLM roleplay	21,117	5556	2694	1385	621	133	0	220	7	8

Table S7. Within-condition variability: conversation-level JSD to the condition mean.

Condition	Speaker	Median JSD	P90 JSD	Outlier frac
HH roleplay	Client	0.451	0.573	0.015
HH roleplay	Counselor	0.488	0.642	0.044
HH real	Client	0.384	0.458	0.038
HH real	Counselor	0.454	0.556	0.000
H-LLM roleplay	Client	0.542	0.620	0.012
H-LLM roleplay	Counselor	0.559	0.641	0.002

Table S8. RQ2 multiplicity control: all thirteen items of the simulated-client battery, comparing the two most recent semesters of the course line (GPT-OSS-120B, $n=23$, vs Llama 3.3 70B, $n=26$). Each row is a two-sided Mann–Whitney test with Cliff’s δ as effect size. Means are on a 5-point disagreement-coded scale (1=agree, 5=disagree), so a negative δ means the GPT-OSS semester was rated more human-like. p_{Holm} is the Holm step-down adjustment across all thirteen items. Only the realism item (A270_02) survives ($p_{\text{Holm}}=.026$). The five acceptance constructs (perceived usefulness, ease of use, enjoyment, behavioral intention, social influence), tested the same way with Holm correction within the construct family, do not separate the semesters (all raw $p \geq .17$, max $|\delta|=0.23$). Rows are ordered by raw p .

Item	Content	$M_{\text{GPT-OSS}}$	M_{Llama}	U	p	p_{Holm}	δ
A270_02	Felt like writing with a real person	2.35	3.42	150	0.002	0.026	-0.50
A270_04	Individual messages were excellent	2.39	3.23	180	0.013	0.155	-0.40
A270_05	Responded understandably to my messages	2.39	2.96	208	0.057	0.632	-0.30
A270_07	Messages coherent and clearly phrased	2.22	2.58	226	0.118	1.000	-0.24
A270_12	No trouble understanding the messages	1.74	1.54	358	0.192	1.000	+0.20
A270_10	Satisfied with the interaction	2.39	2.73	238	0.201	1.000	-0.21
A270_01	Individual messages fit the conversation	2.04	2.35	248	0.259	1.000	-0.17
A270_03	Language quality of the messages	1.87	1.65	345	0.323	1.000	+0.15
A270_13	Reacted adequately to my follow-up questions	2.61	2.85	258	0.393	1.000	-0.14
A270_08	The conversation as a whole was coherent	2.52	2.78	228	0.413	1.000	-0.14
A270_06	Engaged with my questions and suggestions	2.74	2.85	273	0.581	1.000	-0.09
A270_11	Personality felt likeable	2.35	2.23	324	0.599	1.000	+0.09
A270_09	Could run a complete counseling conversation	2.57	2.65	296	0.959	1.000	-0.01

Table S9. Noise-band placement of the two chat comparisons under each unit of analysis, all referenced to HH real’s own split-half band (1,000 splits). “Pctl.” is the empirical percentile of the observed distance within that band. The pattern reported in the main text is unit-invariant: under every unit, and on both roles, both comparisons lie beyond the band.

Unit	Comparison	Role	JSD	Band mean	Band P95	Pctl.	V
SaT spans	HH roleplay	Client	0.233	0.109	0.133	100.0	0.266
	HH roleplay	Counselor	0.242	0.143	0.168	100.0	0.270
	H-LLM	Client	0.453	0.108	0.132	100.0	0.461
	H-LLM	Counselor	0.276	0.143	0.168	100.0	0.262
Regex spans	HH roleplay	Client	0.222	0.107	0.133	100.0	0.253
	HH roleplay	Counselor	0.236	0.154	0.176	100.0	0.268
	H-LLM	Client	0.342	0.107	0.132	100.0	0.291
	H-LLM	Counselor	0.258	0.154	0.178	100.0	0.227
Messages	HH roleplay	Client	0.242	0.138	0.170	100.0	0.276
	HH roleplay	Counselor	0.298	0.173	0.205	100.0	0.330
	H-LLM	Client	0.475	0.137	0.171	100.0	0.501
	H-LLM	Counselor	0.320	0.173	0.205	100.0	0.301

Table S10. The central RQ1 gap restricted to confidently labelled spans.

$\Delta = \text{JSD}(\text{HH real}, \text{H-LLM}) - \text{JSD}(\text{HH real}, \text{HH roleplay})$ is the proximity gap of the main paper with a 95% interval from 20,000 conversation-level bootstrap resamples, and * marks intervals excluding zero. The “all items” row reproduces the SaT row of Table S20, interval included, and its bounds are taken from that run rather than redrawn here. If classifier error generated the effect, discarding the least confidently labelled items would reduce Δ . On the client side it does not, and Δ_{CL} stays between 0.213 and 0.239. On the counselor side Δ_{CO} stays small and its interval includes zero at the 0.90 threshold.

Confidence threshold	Role	Items retained	JSD real-LLM	JSD real-roleplay	Δ [95% CI]
all items	Client	100%	0.453	0.233	+0.220* [0.178, 0.243]
all items	Counselor	100%	0.276	0.242	+0.033* [0.000, 0.049]
≥ 0.70	Client	78%	0.466	0.227	+0.239* [0.191, 0.265]
≥ 0.70	Counselor	77%	0.264	0.223	+0.041* [0.001, 0.059]
≥ 0.90	Client	63%	0.479	0.244	+0.236* [0.189, 0.263]
≥ 0.90	Counselor	62%	0.253	0.210	+0.043 [-0.004, 0.063]
≥ 0.99	Client	42%	0.455	0.242	+0.213* [0.158, 0.245]
≥ 0.99	Counselor	33%	0.265	0.191	+0.075* [0.006, 0.101]

Table S11. Between-condition distance and central gap under each exclusion set, with k the size of the label support after removal and renormalization. Band statistics use HH real’s own split-half band recomputed on the reduced support. Δ carries a 95% interval from 20,000 conversation-level bootstrap resamples, and * marks intervals excluding zero. The “full” row reproduces the SaT row of Table S20, interval included. Its bounds are taken from that run rather than redrawn here, so the baseline is one number and not two independent estimates of it.

Exclusion set	Role	k	JSD real-LLM	JSD real-roleplay	Band mean	Pctl.	Δ [95% CI]
full	Client	38	0.453	0.233	0.109	100.0	+0.220* [0.178, 0.243]
full	Counselor	57	0.276	0.242	0.143	100.0	+0.033* [0.000, 0.049]
S1	Client	33	0.460	0.237	0.108	100.0	+0.223* [0.180, 0.249]
S1	Counselor	51	0.243	0.216	0.148	100.0	+0.027 [-0.013, 0.045]
S1+S2	Client	32	0.391	0.199	0.122	100.0	+0.192* [0.141, 0.219]
S1+S2	Counselor	50	0.267	0.237	0.161	100.0	+0.030 [-0.013, 0.050]
S1+S2+S3	Client	31	0.394	0.191	0.121	100.0	+0.203* [0.151, 0.231]
S1+S2+S3	Counselor	48	0.270	0.239	0.160	100.0	+0.032 [-0.012, 0.052]

Table S12. Numbers behind Figure 2 of the main paper: between-condition JSD anchored in the reference condition’s split-half noise band (1,000 splits), with band mean (noise floor), band P95 (upper edge), the 95th percentile of the size-matched null (nm-P95), and Cramér’s V . All comparisons use HH real as reference.

Comparison	Role	JSD	Band mean	Band P95	nm-P95	V
HH roleplay	Client	0.233	0.109	0.133	0.092	0.27
HH roleplay	Counselor	0.242	0.143	0.168	0.116	0.27
H-LLM	Client	0.453	0.108	0.132	0.073	0.46
H-LLM	Counselor	0.276	0.143	0.168	0.091	0.26
H-LLM [GPT-OSS-120B]	Client	0.361	0.108	0.131	0.087	0.41
H-LLM [GPT-OSS-120B]	Counselor	0.293	0.143	0.167	0.111	0.33
H-LLM [Llama 3.3 70B]	Client	0.524	0.109	0.132	0.076	0.57
H-LLM [Llama 3.3 70B]	Counselor	0.298	0.144	0.169	0.098	0.30
H-LLM [Mixtral 8x7B]	Client	0.495	0.109	0.132	0.086	0.54
H-LLM [Mixtral 8x7B]	Counselor	0.292	0.144	0.170	0.110	0.33

Table S13. Distributional proximity to HH real by model with 95% conversation-level bootstrap CIs (lower = closer). These are the intervals drawn in the per-model rows of Figure 2 of the main paper.

Model	Client JSD [95% CI]	Counselor JSD [95% CI]
GPT-OSS-120B	0.361 [0.336, 0.401]	0.293 [0.274, 0.339]
Llama 3.3 70B	0.524 [0.501, 0.552]	0.298 [0.286, 0.330]
Mixtral 8x7B	0.495 [0.470, 0.529]	0.292 [0.274, 0.338]

Table S14. Sequential structure summary (role-aware transitions) for the three chat conditions. All three are dominated by cross-role alternation. Section 5.3 of the main paper reads these differences.

Condition	Same-role	Cross-role	Self-transition
HH roleplay	0.323	0.677	0.101
HH real	0.365	0.635	0.115
H-LLM roleplay	0.292	0.708	0.070

Table S15. All 70 transition cells that shift significantly for H-LLM vs HH real under FDR control at $q = 0.10$, sorted by absolute shift and set in two column blocks. Rates are row-normalized transition probabilities in each condition. Δ is the H-LLM rate minus the HH real rate, n the combined transition count, and p the empirical p -value from 2,000 split-half resamples of HH real. The smallest attainable value is shown as <0.001 . Section 5.3 of the main paper reports the counts and the largest shifts.

From	To	H-LLM	HH real	Δ	n	p	From	To	H-LLM	HH real	Δ	n	p
CL:Opening	CL:Clarify	0.585	0.192	+0.394	441	<0.001	CL:Clarify	CO:Opening	0.100	0.036	+0.064	421	<0.001
CO:Closing	CL:Help	0.480	0.178	+0.302	260	0.002	CO:Clarify	CL:Resources	0.070	0.008	+0.062	292	<0.001
CO:Resources	CL:Resources	0.337	0.035	+0.302	97	<0.001	CL:Help	CO:Motivation	0.087	0.029	+0.058	156	0.004
CL:Opening	CO:Opening	0.004	0.275	-0.271	36	<0.001	CL:Other	CO:Other	0.048	0.106	-0.058	84	0.011
CO:Objectives	CL:Resources	0.221	0.000	+0.221	146	<0.001	CO:Other	CO:Objectives	0.075	0.020	+0.055	136	<0.001
CO:Moderation	CL:Other	0.025	0.243	-0.218	85	<0.001	CL:Opening	CL:Empathy	0.055	0.000	+0.055	39	<0.001
CO:Help	CL:Clarify	0.175	0.384	-0.209	214	<0.001	CO:Help	CL:Motivation	0.072	0.020	+0.052	61	0.004
CO:Objectives	CL:Objectives	0.227	0.020	+0.206	151	<0.001	CO:Other	CO:Help	0.091	0.042	+0.049	174	0.003
CO:Motivation	CL:Other	0.009	0.203	-0.194	31	0.021	CO:Other	CO:Motivation	0.079	0.032	+0.047	149	0.002
CL:Opening	CL:Other	0.052	0.225	-0.173	64	0.018	CO:Clarify	CL:Motivation	0.081	0.044	+0.037	368	0.015
CO:Other	CL:Clarify	0.031	0.197	-0.165	151	<0.001	CO:Other	CO:Closing	0.068	0.032	+0.036	131	0.004
CO:Help	CL:Help	0.318	0.167	+0.151	284	0.001	CL:Other	CL:Closing	0.040	0.006	+0.034	18	<0.001
CO:Help	CL:Other	0.017	0.157	-0.140	44	0.013	CO:Clarify	CL:Empathy	0.045	0.013	+0.033	196	<0.001
CL:Closing	CL:Help	0.153	0.025	+0.129	36	<0.001	CO:Moderation	CL:Closing	0.036	0.004	+0.032	31	<0.001
CO:Motivation	CL:Resources	0.127	0.000	+0.127	112	<0.001	CL:Other	CL:Motivation	0.054	0.024	+0.030	34	0.012
CO:Help	CL:Resources	0.134	0.010	+0.124	108	<0.001	CL:Resources	CO:Closing	0.030	0.000	+0.030	29	<0.001
CO:Clarify	CL:Other	0.033	0.153	-0.120	266	<0.001	CO:Moderation	CO:Objectives	0.033	0.004	+0.029	29	<0.001
CO:Motivation	CO:Clarify	0.152	0.035	+0.117	138	<0.001	CL:Empathy	CO:Objectives	0.028	0.000	+0.028	10	<0.001
CL:Clarify	CO:Clarify	0.379	0.273	+0.106	1776	0.009	CO:Objectives	CL:Closing	0.027	0.000	+0.027	18	<0.001
CO:Moderation	CL:Objectives	0.113	0.008	+0.105	97	<0.001	CL:Other	CL:Resources	0.031	0.005	+0.027	14	<0.001
CL:Opening	CL:Help	0.151	0.050	+0.101	114	0.008	CL:Help	CO:Objectives	0.035	0.008	+0.026	61	<0.001
CO:Moderation	CL:Help	0.174	0.076	+0.098	166	0.002	CO:Closing	CL:Objectives	0.025	0.000	+0.025	13	<0.001
CO:Other	CL:Other	0.001	0.096	-0.095	50	0.003	CO:Closing	CL:Motivation	0.025	0.000	+0.025	13	<0.001
CL:Objectives	CO:Objectives	0.095	0.000	+0.095	115	<0.001	CO:Moderation	CL:Motivation	0.036	0.011	+0.024	33	0.008
CO:Help	CL:Objectives	0.101	0.010	+0.091	82	<0.001	CO:Opening	CO:Objectives	0.023	0.000	+0.023	22	<0.001
CL:Help	CL:Other	0.012	0.100	-0.088	45	0.024	CO:Other	CO:Resources	0.031	0.010	+0.021	58	0.018
CL:Closing	CO:Moderation	0.110	0.025	+0.085	26	<0.001	CL:Opening	CL:Resources	0.021	0.000	+0.021	15	<0.001
CL:Resources	CO:Objectives	0.083	0.000	+0.083	81	<0.001	CL:Clarify	CL:Objectives	0.034	0.015	+0.019	147	0.019
CL:Clarify	CL:Other	0.013	0.093	-0.079	168	<0.001	CL:Other	CL:Opening	0.029	0.011	+0.017	17	0.028
CL:Motivation	CO:Motivation	0.108	0.029	+0.079	78	0.002	CO:Help	CL:Closing	0.021	0.005	+0.017	18	<0.001
CL:Other	CL:Help	0.105	0.030	+0.075	56	<0.001	CO:Opening	CL:Empathy	0.015	0.000	+0.015	15	<0.001
CO:Clarify	CL:Objectives	0.108	0.037	+0.071	470	<0.001	CL:Help	CL:Resources	0.019	0.004	+0.015	33	<0.001
CL:Help	CO:Closing	0.091	0.025	+0.066	162	<0.001	CO:Motivation	CO:Objectives	0.013	0.000	+0.013	11	<0.001
CL:Other	CL:Objectives	0.083	0.018	+0.065	40	<0.001	CL:Clarify	CL:Resources	0.020	0.008	+0.012	85	0.012
CO:Moderation	CL:Resources	0.064	0.000	+0.064	54	<0.001	CL:Resources	CL:Help	0.010	0.000	+0.010	10	<0.001

Table S16. All 20 transition cells that shift significantly for HH roleplay vs HH real under FDR control at $q = 0.10$, sorted by absolute shift. Columns as in Table S15.

From	To	HH roleplay	HH real	Δ	n	p
CL:Opening	CL:Clarify	0.422	0.192	+0.230	66	<0.001
CO:Closing	CL:Closing	0.287	0.069	+0.219	32	<0.001
CO:Motivation	CO:Clarify	0.232	0.035	+0.196	29	<0.001
CO:Moderation	CL:Other	0.056	0.243	-0.187	78	0.001
CO:Moderation	CL:Clarify	0.422	0.243	+0.178	169	<0.001
CO:Other	CL:Clarify	0.051	0.197	-0.146	114	0.001
CL:Resources	CO:Moderation	0.161	0.036	+0.126	11	0.001
CO:Resources	CL:Resources	0.122	0.035	+0.087	11	0.011
CL:Help	CO:Closing	0.103	0.025	+0.078	22	<0.001
CO:Moderation	CL:Closing	0.072	0.004	+0.069	19	<0.001
CO:Help	CL:Empathy	0.070	0.015	+0.054	11	<0.001
CO:Other	CO:Help	0.085	0.042	+0.043	48	0.007
CO:Other	CO:Objectives	0.060	0.020	+0.040	29	0.003
CL:Clarify	CL:Other	0.054	0.093	-0.039	175	0.016
CO:Moderation	CL:Objectives	0.044	0.008	+0.037	13	<0.001
CL:Other	CL:Objectives	0.049	0.018	+0.032	22	0.001
CO:Other	CO:Closing	0.063	0.032	+0.031	36	0.013
CO:Other	CO:Resources	0.038	0.010	+0.028	17	<0.001
CO:Clarify	CL:Resources	0.025	0.008	+0.017	25	0.009
CL:Clarify	CO:Closing	0.021	0.008	+0.013	32	0.008

Table S17. Message length and surface register of client messages per condition and, within H-LLM, per model, persona brevity cue, and template family (Section 5.2 of the main paper). Median words per client message with interquartile range, and the share of messages without sentence-final punctuation. “Llama x template_family” restricts the template contrast to the one model that ran under both families.

Grouping	Level	Conv.	Client msgs	Median words [IQR]	No final punct.
condition	HH roleplay	68	1366	12 [5, 26]	38.7%
	HH real	53	1632	10 [3, 21]	85.4%
	H-LLM	414	7758	39 [28, 51]	9.2%
model	Mixtral 8x7B	86	1464	30 [22, 39]	0.0%
	Llama 3.3 70B	238	4804	44 [35, 57]	0.3%
	GPT-OSS-120B	90	1490	25 [17, 37]	47.1%
brevity_persona	yes	225	4312	32 [23, 40]	16.5%
	no	189	3446	50 [39, 63]	0.1%
template_family	plain	207	3330	45 [26, 63]	21.5%
	chat	121	2964	39 [33, 47]	0.0%
	not logged	86	1464	30 [22, 39]	0.0%
Llama x template_family	plain	117	1840	59 [47, 72]	0.8%
	chat	121	2964	39 [33, 47]	0.0%

Table S18. Client-side distance to HH real and proximity gap Δ per prompt configuration, SaT spans, with 95% intervals from 20,000 conversation-level bootstrap resamples (Section 5.4 of the main paper). Template family is nested in model and semester: GPT-OSS-120B ran plain templates only and Mixtral 8x7B logged no prompts. The persona-matched block restricts Llama 3.3 70B to the four personas that appear under both families. The difference between its two rows is 0.164 [0.129, 0.194].

Block	Prompt configuration	Conv.	Med. words	JSD real-LLM [95% CI]	Δ [95% CI]
all	all H-LLM	414	38	0.453 [0.429, 0.483]	+0.220* [0.178, 0.243]
template family	plain template	207	52	0.396 [0.372, 0.430]	+0.163* [0.122, 0.190]
	chat template	121	39	0.597 [0.575, 0.625]	+0.364* [0.322, 0.387]
	prompt not logged	86	30	0.495 [0.469, 0.529]	+0.262* [0.220, 0.289]
persona brevity cue	persona asks for brevity	225	30	0.448 [0.422, 0.482]	+0.215* [0.171, 0.242]
	persona silent on length	189	54	0.487 [0.464, 0.517]	+0.254* [0.213, 0.278]
model x template family	Mixtral 8x7B, not logged	86	30	0.495 [0.469, 0.529]	+0.262* [0.220, 0.289]
model x persona brevity cue	Mixtral 8x7B, brevity yes	83	30	0.497 [0.471, 0.531]	+0.264* [0.221, 0.291]
model x template family	Llama 3.3 70B, plain	117	61	0.492 [0.471, 0.521]	+0.259* [0.218, 0.283]
	Llama 3.3 70B, chat	121	39	0.597 [0.575, 0.625]	+0.364* [0.322, 0.387]
model x persona brevity cue	Llama 3.3 70B, brevity yes	70	36	0.600 [0.576, 0.633]	+0.367* [0.324, 0.395]
	Llama 3.3 70B, brevity no	168	56	0.507 [0.485, 0.536]	+0.275* [0.233, 0.299]
model x template family	GPT-OSS-120B, plain	90	26	0.361 [0.336, 0.400]	+0.128* [0.085, 0.161]
model x persona brevity cue	GPT-OSS-120B, brevity yes	72	26	0.362 [0.335, 0.405]	+0.129* [0.083, 0.165]
	GPT-OSS-120B, brevity no	18	29	0.434 [0.414, 0.496]	+0.201* [0.165, 0.255]
Llama, persona-matched	Llama 3.3 70B, plain, persona-matched	45	64	0.491 [0.466, 0.528]	+0.258* [0.215, 0.289]
	Llama 3.3 70B, chat, persona-matched	36	38	0.655 [0.632, 0.686]	+0.422* [0.379, 0.449]

Table S19. Size-matched subsample check (Section 5.5.3 of the main paper). Client-side Δ recomputed on 1,000 random draws of 68 H-LLM conversations without replacement, the size of the HH roleplay condition, pooled and within each model. Columns give the mean, the central 95% range of the draws, and the share of draws above zero.

Pool	Conv. in pool	Mean Δ	[P2.5, P97.5]	Share > 0
all H-LLM	414	+0.225	[0.194, 0.256]	100.0%
Mixtral 8x7B	86	+0.263	[0.254, 0.272]	100.0%
Llama 3.3 70B	238	+0.294	[0.272, 0.316]	100.0%
GPT-OSS-120B	90	+0.130	[0.116, 0.144]	100.0%

Table S20. The central RQ1 gap across units of analysis (Section 5.5 of the main paper). Columns give the client- and counselor-side distances from HH real to H-LLM roleplay and to HH roleplay, and their difference (Δ_{CL} and Δ_{CO}), with a 95% interval from 20,000 conversation-level resamples. * marks a gap whose whole interval lies above zero. The bottom row is a misspecified character-class split, not a real segmentation.

Unit of analysis	Client			Counselor		
	real-LLM	real-roleplay	Δ [95% CI]	real-LLM	real-roleplay	Δ [95% CI]
SaT spans	0.453	0.233	+0.220* [0.178, 0.243]	0.276	0.242	+0.033* [0.000, 0.049]
Regex spans	0.342	0.222	+0.120* [0.086, 0.143]	0.245	0.225	+0.020 [-0.012, 0.041]
Messages	0.475	0.242	+0.233* [0.182, 0.258]	0.320	0.298	+0.022 [-0.022, 0.047]
Naive character split	0.106	0.073	+0.033* [0.011, 0.048]	0.073	0.093	-0.019* [-0.030, -0.010]

Table S21. Prompt replay, all arms (Section 5.7 of the main paper). Client-side JSD to HH real on the span unit and on the message unit with 95% intervals from conversation-level bootstrap resamples (20,000 for spans, 5,000 for messages), Cramér's V , the problem share in the first decile, the pooled solution-space share, the share of conversations with at least one general request and at least one rejection, the negative-to-positive feedback ratio, and the median words per client message.

Arm	JSD spans [95% CI]	JSD messages [95% CI]	V	Problem d1	Solution	Req. conv.	Rej. conv.	Neg:Pos	Words
HH real				0.128	0.045	58%	57%	0.78	10
HH roleplay	0.233 [0.214, 0.278]	0.242 [0.224, 0.298]	0.27	0.114	0.062	54%	32%	0.43	12
O: original prompt, re-classified	0.321 [0.298, 0.360]	0.412 [0.377, 0.466]	0.37	0.210	0.161	39%	24%	0.18	25
A: logged prompt, replayed	0.326 [0.302, 0.365]	0.416 [0.384, 0.467]	0.37	0.184	0.159	47%	22%	0.16	27
B: generic realism prompt	0.341 [0.316, 0.380]	0.453 [0.426, 0.499]	0.39	0.311	0.114	57%	13%	0.14	42
C: measurement-based rules	0.329 [0.307, 0.365]	0.448 [0.421, 0.491]	0.37	0.142	0.054	97%	38%	7.00	12
D: C, two rules dosed	0.233 [0.218, 0.268]	0.281 [0.264, 0.325]	0.26	0.126	0.079	12%	47%	4.59	9
E: D, praise allowed	0.238 [0.224, 0.273]	0.289 [0.270, 0.335]	0.27	0.156	0.083	17%	54%	3.51	9
F: phase-aware rules	0.244 [0.232, 0.278]	0.298 [0.283, 0.341]	0.28	0.160	0.081	2%	57%	1.45	9
G: phase-dependent rules	0.213 [0.200, 0.249]	0.299 [0.284, 0.340]	0.24	0.135	0.057	9%	63%	0.73	9

Table S22. Turn-level phase test under the HH real phase model. For every client turn the original history is decoded before the turn and again with the client's spans appended. Client-led entry is the share of clarification turns after which the state is the help phase, and client-led return is the share of help-phase turns after which the state is the clarification phase. The replay arms share the histories of the GPT-OSS-120B deployment.

Group	Clarification turns	Client-led entry into help	Help-phase turns	Client-led return to clarification
HH real	1319	2.0%	163	17.2%
HH roleplay	1107	4.0%	117	16.2%
GPT-OSS-120B deployment	1087	4.4%	244	10.2%
O	1087	4.3%	244	9.4%
A	1087	4.3%	244	7.0%
B	1087	2.7%	244	14.8%
C	1087	0.6%	244	34.4%
D	1087	0.6%	244	24.6%
E	1087	0.9%	244	21.7%
F	1087	0.5%	244	22.1%
G	1087	0.6%	244	10.2%

Condition and role	Units	Cat.	Words	Acc.	Base	κ	Macro F1	Wtd. F1	Pillar	Group
HH roleplay – all units	4,354	61	10	0.461	0.089	0.445 [0.427, 0.464]	0.344	0.464	0.817	0.621
HH roleplay – counselor	2,467	33	11	0.418	0.158	0.388 [0.367, 0.411]	0.319	0.429	0.778	0.565
HH roleplay – client	1,887	28	9	0.517	0.151	0.481 [0.453, 0.509]	0.381	0.516	0.869	0.694
HH real – all units	3,591	59	10	0.360	0.202	0.339 [0.319, 0.358]	0.304	0.371	0.813	0.623
HH real – counselor	1,727	35	10	0.330	0.142	0.303 [0.277, 0.332]	0.301	0.323	0.765	0.576
HH real – client	1,864	24	9	0.387	0.389	0.332 [0.297, 0.370]	0.313	0.419	0.857	0.666

Table S23. Agreement of the OnCoCo classifier with the expert annotation of the two human chat corpora. Both label sets sit on identical units: the classifier was re-run on the human segments, with the role prefix the pipeline uses, so only the labels differ and not the segmentation. *Units* counts annotated segments, *Cat.* the number of distinct OnCoCo categories the human annotators used in that block (the classifier can choose among 68, of which 66 have a counterpart in the German scheme), and *Words* their median length in whitespace-separated tokens. *Acc.* is unit-level agreement and *Base* the share of the largest human category, the accuracy a constant prediction would reach. κ is Cohen’s kappa with a 95% conversation-level bootstrap interval (2,000 draws). *Macro F1* averages over the categories the humans used, so rare ones weigh as much as frequent ones, while *Wtd. F1* weighs by support. *Pillar* and *Group* are accuracies after truncating both codes to their second and third field, that is to the broad field of action and to the function group within it.

Table S24. Classifier against expert annotation for the categories that carry the signatures of Section 5.2 in the main paper, per human corpus. *Exp.* and *Model* are the shares of that role’s units the experts and the classifier assign to the category, *P*, *R*, and *F1* precision, recall, and F1 of the classifier against the experts. Client categories are shares of client units, counselor categories shares of counselor units. The counselor residual category does not exist in the German scheme of the HH roleplay annotation, so every unit the classifier places there is an error.

Category	HH roleplay					HH real				
	Exp.	Model	P	R	F1	Exp.	Model	P	R	F1
Problem statement	0.051	0.046	0.53	0.49	0.51	0.202	0.068	0.64	0.22	0.32
Problem definition	0.031	0.047	0.26	0.41	0.32	0.010	0.072	0.00	0.03	0.01
Consent	0.065	0.081	0.71	0.88	0.78	0.038	0.063	0.58	0.96	0.72
Rejection	0.011	0.009	0.69	0.57	0.63	0.008	0.014	0.43	0.79	0.56
General request	0.029	0.028	0.78	0.73	0.76	0.022	0.021	0.75	0.71	0.73
Other statements (client)	0.002	0.018	0.08	0.75	0.14	0.009	0.037	0.18	0.74	0.29
Other statements (counselor)	0.000	0.052	0.00	–	0.00	0.008	0.024	0.13	0.38	0.19
Inappropriate remark (counselor)	0.004	0.008	0.03	0.06	0.04	0.005	0.035	0.07	0.47	0.12

Table S25. The three client-side signature buckets of Section 5.2 in the main paper under expert and classifier labels, on the experts’ units. Shares are of client units with 95% conversation-level bootstrap intervals, *Ratio* is the classifier share over the expert share, and *Recall* and *Precision* are computed at bucket level, so a confusion inside the bucket does not count as an error.

Bucket	Condition	Expert share [95% CI]	Model share [95% CI]	Ratio	Recall	Precision
Problem disclosure	HH roleplay	0.187 [0.170, 0.207]	0.215 [0.195, 0.235]	1.15	0.71	0.62
Problem disclosure	HH real	0.405 [0.366, 0.445]	0.268 [0.233, 0.306]	0.66	0.47	0.71
Solution-space engagement	HH roleplay	0.101 [0.085, 0.116]	0.079 [0.066, 0.092]	0.78	0.46	0.59
Solution-space engagement	HH real	0.062 [0.044, 0.085]	0.069 [0.055, 0.083]	1.11	0.33	0.30
Reciprocal requests	HH roleplay	0.068 [0.050, 0.087]	0.064 [0.049, 0.079]	0.94	0.73	0.78
Reciprocal requests	HH real	0.043 [0.028, 0.061]	0.040 [0.027, 0.055]	0.94	0.71	0.76

Table S26. Client-side proximity gap under equal conversation weighting and persona matching. JSD to HH real for H–LLM and HH roleplay, and their difference Δ_{CL} with a 95% conversation-level bootstrap interval. The first row reproduces the pooled result of the main paper.

Variant	HH roleplay conv.	H–LLM conv.	JSD(real, H–LLM)	JSD(real, HH roleplay)	Δ_{CL} [95% CI]
pooled spans (paper)	68	414	0.453	0.233	+0.220 [+0.178, +0.243]
equal-weighted conversations	68	414	0.442	0.225	+0.217 [+0.174, +0.239]
persona-matched, pooled spans	40	220	0.458	0.251	+0.208 [+0.160, +0.232]
persona-matched, equal-weighted	40	220	0.450	0.241	+0.209 [+0.159, +0.234]

Table S27. Conversation-resampled and student-resampled 95% intervals for the proximity gaps and the per-model distances to HH real. Students are the platform’s course-member accounts within each export. HH real and HH roleplay are resampled by conversation in both columns.

Quantity	Conv.	Students	Estimate	95% CI, conversations	95% CI, students
Δ_{CL}	414	281	+0.220	[+0.178, +0.243]	[+0.177, +0.245]
JSD to HH real, GPT-OSS-120B, Client	90	64	0.361	[0.337, 0.402]	[0.338, 0.401]
JSD to HH real, Llama 3.3 70B, Client	238	160	0.524	[0.502, 0.553]	[0.500, 0.553]
JSD to HH real, Mixtral 8x7B, Client	86	57	0.495	[0.470, 0.529]	[0.468, 0.533]
Δ_{CO}	414	281	+0.033	[+0.000, +0.049]	[+0.000, +0.050]
JSD to HH real, GPT-OSS-120B, Counselor	90	64	0.293	[0.274, 0.338]	[0.273, 0.339]
JSD to HH real, Llama 3.3 70B, Counselor	238	160	0.298	[0.285, 0.330]	[0.285, 0.331]
JSD to HH real, Mixtral 8x7B, Counselor	86	57	0.292	[0.275, 0.337]	[0.275, 0.340]

Table S28. Client-side signature shares by absolute client turn, aligned at the start and at the end of the conversation. A client turn is a maximal run of consecutive client messages. Shares are of all client spans in the bin, pooled over conversations. Problem disclosure is *Problem statement* and *Problem definition*. Solution space is the feedback on recommendations, the report on their implementation, and the two resource-activation labels, as in Section 5.2 of the main paper.

Aligned at	Client turn	Problem disclosure			Solution space			Rejection		
		HH real	HH roleplay	H–LLM	HH real	HH roleplay	H–LLM	HH real	HH roleplay	H–LLM
Start	1	0.020	0.116	0.226	0.000	0.007	0.012	0.000	0.007	0.002
	2	0.163	0.222	0.357	0.022	0.009	0.033	0.000	0.009	0.000
	3	0.195	0.200	0.445	0.034	0.011	0.028	0.011	0.032	0.006
	4–6	0.335	0.347	0.333	0.033	0.030	0.096	0.033	0.023	0.007
	7–10	0.246	0.257	0.199	0.058	0.051	0.204	0.023	0.033	0.005
	11–15	0.203	0.153	0.107	0.050	0.098	0.310	0.040	0.022	0.007
	16–20	0.208	0.090	0.106	0.056	0.064	0.358	0.023	0.021	0.007
	21+	0.091	0.080	0.077	0.044	0.108	0.404	0.036	0.019	0.005
End	last	0.039	0.000	0.014	0.000	0.021	0.345	0.008	0.000	0.002
	2–3 from end	0.027	0.044	0.032	0.048	0.133	0.380	0.016	0.006	0.012
	4–6 from end	0.134	0.098	0.102	0.086	0.109	0.313	0.034	0.025	0.005
	7–10 from end	0.190	0.198	0.221	0.051	0.096	0.207	0.045	0.029	0.008

Table S29. Held-out prompt replay on the 117 Llama 3.3 70B conversations of the plain-text template family, client side. Arms A, D, and G are generated by GPT-OSS-120B on the original histories, and arm O is the original Llama text through the current pipeline. Columns as in Table 4 of the main paper, plus the JSD to HH roleplay. Intervals are conversation-level bootstraps with 20,000 resamples.

Group	JSD to HH real [95% CI]	Δ [95% CI]	JSD to roleplay	Problem d1	Solution	Rej. conv.	Neg:Pos	Words	Entry
HH real				0.128	0.045	57%	0.78	10	2.0%
HH roleplay	0.233 [0.214, 0.278]	+0.000 [−0.035, +0.034]		0.114	0.062	32%	0.43	12	4.0%
Llama 3.3 70B deployment (plain template)	0.492 [0.471, 0.521]	+0.259 [+0.218, +0.283]	0.398	0.163	0.149	8%	0.05	59	3.2%
O: original text, re-classified	0.462 [0.441, 0.492]	+0.229 [+0.189, +0.254]	0.369	0.193	0.142	8%	0.05	59	3.2%
A: logged prompt, GPT-OSS	0.421 [0.401, 0.452]	+0.188 [+0.146, +0.216]	0.359	0.136	0.062	14%	0.18	26	5.0%
D: dosed rules, unchanged	0.287 [0.270, 0.321]	+0.054 [+0.010, +0.089]	0.327	0.074	0.057	51%	7.32	8	1.6%
G: phase-dependent rules, unchanged	0.253 [0.236, 0.287]	+0.020 [−0.024, +0.056]	0.309	0.095	0.047	68%	2.15	7	0.7%

Table S30. Pipeline and generation parameters.

Stage	Setting
Segmentation	SaT-6l (<code>segment-any-text/sat-6l</code>) with a LoRA adapter (rank 128, alpha 256, dropout 0.1, attention query and value), boundary threshold 0.5, sliding windows of 512 tokens. wtpsplit 2.1.7, adapters 1.2.0.
Classification	XLm-RoBERTa-large fine-tuned on OnCoCo 1.0, the study checkpoint (68 output units, two of them counselor codes outside the 66-category scheme that are never predicted in the corpora), input prefixed with “Counselor:” or “Client:”, no role masking of the output, batch size 16. transformers 4.51.3, torch 2.10.0.
Distance Intervals	Jensen–Shannon distance, square root of the base-2 divergence, over pooled span counts per condition and role. Percentile bootstrap over conversations, seed 42. 20,000 resamples for proximity gaps, per-model distances, student-clustered intervals, and replay arms on spans. 5,000 for replay arms on messages, weighting checks, and absolute turn positions.
Noise scale	Split-half band from 1,000 random halvings of HH real. Size-matched null from 1,000 draws with replacement at the compared sizes.
Transitions	2,000 split-halves of HH real, two-sided empirical p for off-diagonal cells with at least 10 transitions, Benjamini–Hochberg at $q = 0.10$.
Phase model	Categorical HMM (hmmlearn 0.3.3), three states, 20 random initialisations with seeds 42 to 61, at most 500 EM iterations, tolerance 10^{-4} , best log-likelihood kept.
Deployments	Logged sampling of the GPT-OSS-120B and Llama 3.3 70B deployments on a hosted inference service: temperature 1, at most 3,000 tokens, no frequency or presence penalty. The Mixtral 8x7B deployment logged neither prompts nor settings.
Replay	<code>openai/gpt-oss-120b</code> served with vLLM 0.28 on one A100 80 GB, temperature 1, at most 3,000 tokens, no penalties, top- p and reasoning effort at server defaults, no fixed seed, prompt sent as one user message, context length 16,384 tokens.

Table S31. Sentiment of client messages per condition. Shares of messages that `oliverguhr/german-sentiment-bert` classifies as negative, neutral, or positive.

Condition	Client messages	Negative	Neutral	Positive
HH real	1,632	0.586	0.170	0.244
HH roleplay	1,366	0.455	0.220	0.324
H-LLM	7,758	0.451	0.196	0.353